

[25/06/2026 11:40 PM] hssnd: Dr. Doc:
Welcome to Embryology of Cardiovascular System 😊

Today we'll cover...

Early Heart Development

- ✓ Primordial Formation
- ✓ Fusion and initial positioning
- ✓ Layers of the Primitive Heart Wall
- ✓ Positional shifts
- ✓ Heart tube elongation
- ✓ Cardiac Looping

Early Heart Development

🌀 The heart is the first functional organ to develop in the embryo. Development begins during the 3rd week and the primitive heart starts beating by approximately day 22–23.

1. Primordial Formation (Middle of 3rd Week)

Formation of Angioblastic Cords

During gastrulation, mesoderm differentiates into:

- ✓ Paraxial mesoderm
- ✓ Intermediate mesoderm
- ✓ Lateral plate mesoderm

🌀 The lateral plate mesoderm splits into:

1. Somatic (parietal) mesoderm
2. Splanchnic (visceral) mesoderm

👉 The heart develops from the splanchnic mesoderm located cranial to the oropharyngeal membrane in the cardiogenic area.

🌀 Process

✳️ Cells of the splanchnic mesoderm proliferate and form clusters called angioblastic cell clusters.

✳️ These clusters elongate into angioblastic cords.

✳️ The cords later canalize to form vascular channels.

🌀 Importance

These angioblastic cords are the earliest precursors of:

- ✳️ Endocardium
- ✳️ Primitive heart tubes
- ✳️ Great vessels

🌀 Canalization into Paired Endocardial Heart Tubes

🌀 The angioblastic cords develop small cavities within them.

👉 This process is called canalization.

🌀 The cavities fuse to form:

- ✓ Right endocardial heart tube

- ✓ Left endocardial heart tube
 - ☞ These paired tubes lie within the cardiogenic region.

📌 Features

- ▼ Each tube is lined by endothelial cells.
- ▼ The tubes are surrounded by splanchnic mesoderm.
- ▼ Initially, they are bilateral structures.

2. Fusion and Initial Positioning

📌 Cephalocaudal Fusion After Lateral Folding (Days 21–22)

As the embryo undergoes folding:

A. Lateral Folding

- ✧ The two endocardial tubes move toward the midline.
- ✧ They fuse into a single primitive heart tube.

B. Cephalocaudal Folding

- 🌀 Rapid growth of the brain causes the cardiogenic area to move:
 - ✧ From a cranial position to a ventral and caudal position
- 👤 Thus, the heart comes to lie:
 - ✓ Ventral to the foregut
 - ✓ Caudal to the oropharyngeal membrane

📌 Primitive Heart Tube

🌀 After fusion, the primitive heart tube consists of:

1. Endocardial tube internally
2. Myocardial layer externally
3. Cardiac jelly between them

✧ The heart tube begins rhythmic contractions around day 22.

📌 Dorsal Mesocardium

- ✓ Initially, the heart tube is suspended within the pericardial cavity by a fold of mesoderm called the dorsal mesocardium
- ✓ It attaches the heart tube to the dorsal wall of the pericardial cavity.

Functions

- ✧ Stabilizes the primitive heart tube
- ✧ Carries blood vessels and nerves

📌 Formation of the Transverse Pericardial Sinus

- ✓ The central portion of the dorsal mesocardium degenerates.
- ✓ This creates a communication between the right and left sides of the pericardial cavity.
 - ☞ The resulting passage is called the Transverse Pericardial Sinus.

📌 Clinical Importance

- 👤 In adult cardiac surgery, surgeons pass clamps through this sinus to control:
 - ✓ Ascending aorta

✓ Pulmonary trunk

3. Layers of the Primitive Heart Wall

✱ The primitive heart wall develops four important layers.

▼ A. Endocardium

🕒 Origin: Derived from endothelial lining of the fused endocardial tubes.

🕒 Structure

✓ Innermost layer

✓ Thin endothelial lining

🕒 Functions

✱ Forms endothelial lining of heart chambers

✱ Contributes to heart valves

✱ Participates in septation

▼ B. Primordial Myocardium

🕒 Origin: Derived from surrounding splanchnic mesoderm.

🕒 Structure

✱ Thick muscular layer around endocardium

🕒 Functions

✱ Responsible for cardiac contractions

✱ Forms the muscular wall of the heart

👉 Initially, contractions are weak and peristaltic.

▼ C. Cardiac Jelly

✱ Located between endocardium and myocardium.

[25/06/2026 11:40 PM] hssnd: 🕒 Composition

A gelatinous extracellular matrix rich in:

✓ Glycosaminoglycans

✓ Proteoglycans

🕒 Functions

Cardiac jelly is extremely important because it:

👉 Acts as primitive connective tissue

👉 Participates in formation of:

✓ Endocardial cushions

✓ Atrioventricular septa

✓ Heart valves

🕒 Clinical Importance

Abnormal cardiac jelly development may cause:

✱ Septal defects

✱ Valve malformations

▼ D. Epicardium (Visceral Pericardium)

Origin: Derived from mesothelial cells from the:

- ✓ Proepicardial organ
- ✓ Septum transversum region

Development

- ✓ Mesothelial cells spread over the myocardium to form the Epicardium.

📌 Functions

👉 Forms visceral layer of serous pericardium

Gives rise to:

- ✳️ Coronary vessels
- ✳️ Cardiac connective tissue

4. Positional Shifts (Head Folding)

Repositioning of the Heart Ventral to the Foregut

🕒 Originally, the cardiogenic area lies cranial to the neural plate

🕒 After head folding:

- ✓ The heart shifts ventrally
- ✓ Comes to lie anterior to the foregut
- 👉 This establishes the future thoracic position.

📌 Reversal of Heart Tube and Pericardium Relation

✳️ Initially pericardial cavity lies dorsal to the heart-forming region.

🕒 After folding:

- ✓ Heart becomes ventral to the pericardial cavity.
- 👉 This reversal is due to embryonic folding movements.

📌 Relation with the Septum Transversum

✓ The septum transversum is a thick mesodermal plate that later contributes to Central tendon of diaphragm.

🕒 After folding:

- ✓ The heart lies cranial to the septum transversum.
- ✓ The liver develops caudal to it.
- 👉 This relationship is important for future thoracic and abdominal organization.

5. Heart Tube Elongation

🕒 The primitive heart tube elongates rapidly because cells are continuously added from the Secondary heart field.

🕒 The tube cannot remain straight within the confined pericardial cavity.

👉 Therefore, it bends and enlarges.

▼ A. Truncus Arteriosus

- ✓ Most cranial part.

Functions

- ✓ Outflow tract of primitive heart

Derivatives

Forms:

- ↗ Ascending aorta
- ↗ Pulmonary trunk

▼ B. Bulbus Cordis

🌀 Located caudal to truncus arteriosus.

Divisions

1. Proximal bulbus cordis
→ Trabeculated right ventricle
2. Conus cordis
→ Outflow tracts of ventricles

▼ C. Primitive Ventricle

-Forms Trabeculated part of left ventricle

D. Primitive Atrium

↗ Develops into Trabeculated portions of both atria

Including:

✳ Auricles

▼ E. Sinus Venosus

✓ Most caudal part.

✓ Receives all venous blood of the embryo.

Right and Left Sinus Horns

🌀 Each horn receives three paired veins:

1. Vitelline veins
→ From yolk sac
2. Umbilical veins
→ From placenta
3. Common cardinal veins
→ From embryo body

Fate of Sinus Horns

✳ Right Horn Becomes incorporated into right atrium forming Smooth-walled sinus venarum.

✳ Left Horn Forms:

- Coronary sinus
- Oblique vein of left atrium

6. Cardiac Looping

🌀 Cardiac looping is essential for establishing the final spatial arrangement of heart chambers.

↗ It begins around day 23.

Formation of S-Shaped Loop

As elongation continues:

- ✓ Bulbus cordis bends ventrally, caudally, and to the right.
- ✓ Primitive ventricle shifts leftward.

👉 This creates an S-shaped heart tube

🕒 Formation of U-Shaped Bulboventricular Loop

🕒 By the end of the 4th week The bulboventricular region forms a U-shaped loop.

✓ This process is called Dextro-looping (D-looping) because the loop bends toward the right side.

🕒 Positional Changes During Looping

Ventricular Region

Moves:

→Ventrally

→Caudally

→To the right

Atrial Region and Sinus Venosus

Move:

→Dorsally

→Cranially

→To the left

✳ Eventually, the atria come to lie posterior to the ventricles.

🕒 Significance of Cardiac Looping

Looping establishes:

✳ Right-left asymmetry

✳ Correct chamber alignment

✳ Proper arterial and venous connections

🕒 Clinical Correlation: Abnormal Looping

🕒 Dextrocardia

→Heart loops abnormally to the left instead of right.

→Heart lies on right side of thorax.

May occur with:

✳ Situs inversus

🕒 Congenital Malformations from Looping Errors

Abnormal looping can cause:

✓ Transposition of great arteries

✓ Double outlet right ventricle

✓ Septation defects

[25/06/2026 11:40 PM] hssnd: Today we'll cover...

Heart Partitioning and Internal Development

✓ Atrioventricular(AV) Canal

✓ Primordial Atrium(Atrial Septation)

✓ Ventricular Septum

✓ Bulbus Cordis and Outflow Tract

✓ Valves and Conducting system

✓ Development of Conducting System

Heart Partitioning and Internal Development

👁 Overview of Cardiac Septation

🌀 Partitioning of the primitive heart converts the single heart tube into a four-chambered heart with separate pulmonary and systemic circulations.

🌀 This process mainly occurs between the 4th and 8th weeks of development and involves:

1. Partitioning of the atrioventricular canal
2. Septation of the atria
3. Septation of the ventricles
4. Division of the outflow tract
5. Development of valves and conducting tissue

1. Atrioventricular (AV) Canal

👁 Formation of Endocardial Cushions

✓ The primitive heart initially contains a single common atrioventricular canal connecting the primitive atrium and primitive ventricle.

✓ Two localized swellings appear in the AV canal: Dorsal (superior) endocardial cushion and Ventral (inferior) endocardial cushion.

✓ These cushions are formed by proliferation of mesenchymal cells, migration of extracellular matrix (cardiac jelly), and endothelial-to-mesenchymal transformation.

👁 Functions of Endocardial Cushions

✳️ They are very important because they contribute to division of AV canal, formation of AV septum, formation of AV valves, and membranous part of interventricular septum.

👁 Fusion of Cushions (5th Week)

✓ During the 5th week, dorsal and ventral cushions grow toward each other and fuse.

✓ The common AV canal becomes divided into Right AV canal (future tricuspid opening) and Left AV canal (future mitral opening)

👉 This fused structure is called the Septum Intermedium

✳️ It partially separates atria from ventricles.

👁 Clinical Correlation

Failure of cushion fusion causes:

✳️ Atrioventricular septal defects (AVSD)

✳️ Persistent common AV canal

→ Common in Down syndrome (Trisomy 21)

2. Primordial Atrium (Atrial Septation)

✓ Initially, the primitive atrium is a single chamber.

✓ Atrial septation occurs through formation of septum primum, septum secundum, and foramen ovale.

▼ A. Septum Primum and Foramen Primum

👁 Formation of Septum Primum

✓ A thin crescent-shaped membrane called the septum primum grows downward from the roof of the primitive atrium toward the endocardial cushions.

✱ Initially, a gap remains between the free edge of septum primum and endocardial cushions.

👉 This opening is Foramen Primum

🕒 It allows blood flow from right atrium to left atrium in fetal life.

🕒 Closure of Foramen Primum

✓ As septum primum continues growing, foramen primum becomes progressively smaller.

👉 Eventually septum primum fuses with endocardial cushions and foramen primum closes.

▼ B. Formation of Foramen Secundum

🕒 Before foramen primum completely closes Programmed cell death (apoptosis) creates perforations in the upper part of septum primum.

🕒 These perforations merge to form Foramen Secundum

✱ It maintains right-to-left shunting of blood.

▼ C. Formation of Septum Secundum

✱ A second thicker muscular septum develops to the right of septum primum.

-This is Septum Secundum

🕒 It grows downward but never completely closes.

✱ An opening remains called Foramen Ovale

D. Valve of Foramen Ovale

✓ The remaining lower part of septum primum acts as a Flap Valve of Foramen Ovale

Function

✱ Blood flows from right atrium to left atrium, but reverse flow is prevented.

✓ This is essential in fetal circulation because lungs are nonfunctional before birth.

Closure After Birth

🕒 At birth lungs expand, pulmonary circulation increases, left atrial pressure becomes greater than right atrial pressure.

✓ This presses septum primum against septum secundum.

→ Functional closure occurs immediately.

🕒 Later anatomical fusion forms the fossa ovalis.

🕒 Clinical Correlations

✱ Patent Foramen Ovale (PFO)

→ Incomplete fusion after birth.

✱ Atrial Septal Defects (ASD)

→ Common types:

ostium secundum defect, ostium primum defect.

3. Ventricular Septum

[25/06/2026 11:40 PM] hssnd: ✨ Initially, primitive ventricles communicate freely.
Septation forms:

- 1.muscular IV septum
- 2.membranous IV septum.

▼ A. Muscular Interventricular Septum

🌀 A muscular ridge grows upward from the floor of primitive ventricle.

✓ This forms Muscular IV Septum

✓ It enlarges toward endocardial cushions. However, it does not completely close the interventricular communication.

✨ A temporary opening remains Interventricular (IV) Foramen

▼ B. Closure of IV Foramen (End of 7th Week)

🌀 The IV foramen closes by fusion of right bulbar ridge, left bulbar ridge, endocardial cushions, muscular IV septum.

✓ This forms the Membranous Part of IV Septum

✓ Closure occurs near the end of the 7th week.

👁 Clinical Correlation

✨ Ventricular Septal Defect (VSD)

✓ Most common congenital heart defect.

✓ Usually involves membranous septum.

4. Bulbus Cordis and Outflow Tract

🌀 The bulbus cordis contributes to formation of ventricular outflow regions.
It has three important parts:

- 1.proximal part
- 2.conus cordis
- 3.truncus arteriosus.

A. Incorporation of Bulbus Cordis

▼ Right Side

✨ Proximal bulbus cordis becomes incorporated into right ventricle forming Conus Arteriosus (Infundibulum)

✓ Smooth outflow tract of right ventricle leading to pulmonary trunk.

▼ Left Side

✨ Conus cordis contributes to smooth outflow tract of left ventricle Aortic Vestibule

B. Neural Crest Cell Migration

🌀 Neural crest cells migrate into:
truncus arteriosus and
conus cordis.

They are essential for:

✓ septation of outflow tract

- ✓ formation of aorticopulmonary septum.

C. Formation of Aorticopulmonary Septum

✧ Inside truncus arteriosus truncal ridges and bulbar ridges form.

✧ These ridges grow toward each other, twist spirally, and fuse.

✓ This forms the Aorticopulmonary Septum

✓ It divides the outflow tract into ascending aorta and pulmonary trunk

Spiral Nature

✧ The septum spirals approximately 180°.

✓ This explains why pulmonary trunk connects to right ventricle, and aorta connects to left ventricle.

👁 Clinical Correlations

✧ Abnormal neural crest migration causes:

- ✓ Tetralogy of Fallot
- ✓ Persistent truncus arteriosus
- ✓ Transposition of great arteries.

5. Valves and Conducting System

A. Development of AV Valves

✧ The AV valves arise from subendocardial tissue swellings around AV canal.

▼ Process

1. Mesenchymal tissue proliferates.
2. Ventricular myocardium excavates tissue.
3. Valve cusps become attached to papillary muscles via chordae tendineae.

Final AV Valves

✧ Right Side forms Tricuspid valve and Left Side forms Mitral valve

B. Formation of Semilunar Valves

✧ Semilunar valves develop from truncal swellings, and bulbar ridges.

- ✓ Three tubercles form around each arterial opening.
- ✓ These are sculpted into thin cusps.

Final Semilunar Valves

- ✓ Aortic Valve-between left ventricle and aorta.
- ✓ Pulmonary Valve-between right ventricle and pulmonary trunk.

6. Development of Conducting System

🌀 The conducting system develops from specialized cardiomyocytes.

▼ A. Sinoatrial (SA) Node

- ✓ The SA node develops in the right wall of sinus venosus.
- ✓ As the sinus venosus becomes incorporated into right atrium SA node shifts to superior right atrial wall near SVC opening.

Function

- ✧ Becomes the primary pacemaker of the heart.

B. Atrioventricular (AV) Node and Bundle

- ✧ The AV node and AV bundle develop from cells of AV canal region, and left sinoatrial valve tissue.
- ✓ They form specialized conducting fibers connecting atria and ventricles.

Today we'll cover...

Arterial and Venous Development

- ✧ Pharyngeal Arch Arteries
- ✧ Fetal Arterial Fates
- ✧ Venous Circuits

Fetal and Neonatal Circulation

- ✓ Fetal Dynamics
- ✓ Postnatal Changes

Congenital Anomalies

- 🌀 Positional Defects
- 🌀 Septal Defects
- 🌀 Complex Malformations
- 🌀 Vascular Anomalies

Arterial and Venous Development

The development of the cardiovascular system involves the complex transformation of primordial vessels into the final fetal arrangement. This process is driven by vasculogenesis (formation of new vessels from angioblasts) and angiogenesis (budding and branching from existing vessels).

1. Pharyngeal Arch Arteries

[25/06/2026 11:40 PM] hssnd: During the fourth and fifth weeks, six pairs of pharyngeal arch arteries arise from the aortic sac to supply the developing pharyngeal arches. By the eighth week, these are transformed into the fetal arterial pattern:

🌀 1st Pair: Mostly disappear; remnants form parts of the maxillary arteries and may contribute to external carotid arteries.

🌀 2nd Pair: Dorsal parts persist to form the stapedial arteries in the middle ear.

🌀 3rd Pair: Proximal parts form the common carotid arteries; distal parts join the dorsal aortae to form internal carotid arteries.

🌀 4th Pair: The left artery forms part of the arch of the aorta; the right forms the proximal part of the right subclavian artery.

🌀 5th Pair: These are rudimentary and either degenerate or do not develop at all.

🌀 6th Pair: The proximal parts form the proximal pulmonary arteries on both sides. On the left, the distal part forms the ductus arteriosus, a prenatal shunt. On the right, the distal part degenerates.

Relation to Recurrent Laryngeal Nerves: The transformation of the sixth arch arteries explains the differing courses of these nerves. Originally, they hook around the sixth arch arteries. On the left, the nerve remains hooked around the derivative (ductus arteriosus/ligamentum arteriosum). On the right, because the distal sixth arch disappears, the nerve moves cranially to hook around the fourth arch derivative (right subclavian artery).

2. Fetal Arterial Fates

The dorsal aortae give rise to the arteries for the rest of the body through three sets of branches:

✧ Dorsolateral (Somatic Intersegmental) Branches: Form the arteries of the limbs, as well as the intercostal, lumbar, and lateral sacral arteries.

✧ Lateral Splanchnic Branches: Develop into the phrenic, suprarenal, renal, and gonadal vessels.

✧ Ventral Splanchnic Branches: Give rise to the vitelline and umbilical arteries, as well as arteries for the bronchi and esophagus.

✓ Vitelline Arteries: These become the celiac, superior mesenteric (SMA), and inferior mesenteric (IMA) arteries.

✓ Umbilical Arteries: These arise from the ventral splanchnic branches and carry blood to the placenta.

3. Venous Circuits

Venous development follows a cephalocaudal (head-to-tail) direction and involves three major circuits: vitelline, umbilical, and cardinal.

🌀 Vitelline Veins

✧ Pre-hepatic: Form an anastomosis around the duodenum that becomes the portal vein.

✧ Hepatic: Interrupted by liver cords to form hepatic sinusoids.

✧ Post-hepatic: The left disappears, while the right forms the hepatic veins and the hepatic segment of the IVC.

🌀 Umbilical Veins: The right vein disappears, but the left umbilical vein persists to carry oxygenated blood from the placenta. A shunt called the ductus venosus develops through the liver to connect the left umbilical vein directly to the IVC. After birth, the left umbilical vein becomes the ligamentum teres, and the ductus venosus becomes the ligamentum venosum.

☞ Cardinal Veins: These are the main drainage system of the embryo.

✧ Anterior Cardinals: Their cranial parts form the internal jugular veins. An oblique anastomosis between them becomes the left brachiocephalic vein, while the middle part of the right vein forms the right brachiocephalic.

✧ Posterior Cardinals: These largely disappear with the mesonephroi (early kidneys). Their only adult derivatives are the root of the azygos vein and the common iliac veins.

✧ Subcardinal/Supracardinal Replacements: These new veins replace the posterior cardinals. Subcardinal veins form the stem of the left renal vein, suprarenal veins, gonadal veins, and the prerenal IVC segment. Supracardinal veins (cranial to the kidney) form the azygos and hemiazygos veins; the right supracardinal (caudal to the kidney) forms the postrenal segment of the IVC.

☞ Venae Cavae Development

[25/06/2026 11:40 PM] hssnd: The venae cavae are formed by the fusion of various primordial venous segments:

✧ Superior Vena Cava (SVC): Formed from the caudal part of the right anterior cardinal vein and the right common cardinal vein.

✧ Inferior Vena Cava (IVC): Composed of four distinct segments:

- 1) Hepatic segment: From the right vitelline vein.
- 2) Prerenal segment: From the right subcardinal vein.
- 3) Renal segment: From the subcardinal-supracardinal anastomosis.
- 4) Postrenal segment: From the right supracardinal vein.

Fetal and Neonatal Circulation

The transition from fetal to neonatal circulation involves the closure of specialized shunts that redirected blood flow while the lungs were nonfunctional.

I. Fetal Dynamics

In the fetus, the lungs are nonfunctioning, and the placenta serves as the organ for gas exchange. Oxygenated blood is brought from the placenta by the left umbilical vein. Because the lungs do not yet require high blood flow, several shunts allow blood to bypass them and mix with the systemic circulation:

☞ Mixing of Blood: In the fetal state, oxygenated and deoxygenated blood mix at several key locations:

☞ The Liver and IVC: The ductus venosus develops through the liver to connect the left umbilical vein directly with the inferior vena cava (IVC). This allows oxygenated placental blood to mix with deoxygenated blood returning from the lower body.

☞ Right and Left Atria: Once blood enters the right atrium, a significant portion is shunted directly to the left atrium through the oval foramen (foramen ovale).

☞ Descending Aorta: The ductus arteriosus acts as a prenatal shunt, carrying blood from the left pulmonary artery to the dorsal (descending) aorta, effectively bypassing the

pulmonary circulation.

II. Postnatal Changes

Upon birth, the baby's first breath and the subsequent inflation of the lungs lead to immediate physiological shifts that result in the anatomical closure of these fetal shunts.

🔄 Closure of Fetal Shunts: Over time, these temporary vessels and openings transform into adult ligamentous structures:

✳️ Foramen Ovale: After birth, the valve of the oval foramen fuses with the septum secundum, and the opening becomes the oval fossa (fossa ovalis). In approximately 25% of the population, this fusion is incomplete, resulting in a "probe patent" foramen ovale, which is usually not clinically significant.

✳️ Ductus Arteriosus: This shunt constricts and eventually obliterates to become the ligamentum arteriosum.

✳️ Ductus Venosus: This wide channel through the liver obliterates after birth to form the ligamentum venosum.

✳️ Left Umbilical Vein: The persistent left umbilical vein, which carried oxygenated blood during fetal life, obliterates to form the ligamentum teres (round ligament) of the liver. While these changes are typically permanent, certain anomalies such as patent ductus arteriosus or atrial septal defects (ASD) occur if these shunts fail to close properly after birth.

Congenital Anomalies

Congenital cardiovascular malformations are the largest category of human birth defects, accounting for approximately 20% of all congenital defects observed in live-born infants. Most of these abnormalities occur during the critical developmental window between the third and eighth weeks of gestation.

👁️ 1. Positional Defects

✳️ Dextrocardia: is a rare congenital condition in which the heart is located on the right side of the chest instead of the left. Some people have no symptoms, while others may have associated heart defects. It is usually diagnosed through imaging tests and may not require treatment if no other abnormalities are present.

✳️ Ectopia cordis: a rare positional defect where the heart lies exposed on the surface of the thorax. This occurs due to the failure of the sternal plates to fuse or faulty development of the sternum and pericardium during the fourth week.

👁️ 2. Septal Defects

Septal defects involve openings in the walls separating the heart chambers, leading to various degrees of blood shunting.

[25/06/2026 11:40 PM] hssnd: 🌀 A. Atrial Septal Defects (ASD): ASDs are more common in females than males.

✧ Ostium Secundum: This is a significant defect characterized by a large opening between the atria. It is caused by excessive resorption of the septum primum or inadequate development of the septum secundum.

✧ Ostium Primum: This occurs when the septum primum fails to fuse with the endocardial cushions, often resulting in a patent foramen primum and a cleft in the anterior cusp of the mitral valve.

✧ Sinus Venosus: These rare "high ASDs" are located in the superior part of the interatrial septum near the entrance of the superior vena cava (SVC). They result from incomplete absorption of the sinus venosus into the right atrium or abnormal septum secundum development.

✧ Common Atrium (Cor Triloculare Biventriculare): This rare condition involves the complete absence of the interatrial septum due to the total failure of the septum primum and septum secundum to develop.

✧ Patent Foramen Ovale (PFO): While not always a clinical "defect," a probe-patent oval foramen is present in up to 25% of people and results from incomplete adhesion between the valve of the foramen ovale and the septum secundum after birth.

🌀 B. Ventricular Septal Defects (VSD):-VSD is the most common congenital anomaly of the heart.

✧ Membranous VSD: This is the most frequent type, caused by the failure of the right and left bulbar ridges to fuse with the AV cushions. This results in a left-to-right shunt of blood, reducing systemic output and causing fatigue.

✧ Muscular VSD: The muscular part of the interventricular septum are rare.

3. Complex Malformations

🌀 Tetralogy of Fallot: This is the most common congenital cyanotic heart disease. It consists of four classic malformations:

1. Pulmonary stenosis (narrowing of the pulmonary trunk).
2. Overriding aorta (the aortic orifice is displaced to the right to override the IV septum).
3. Interventricular septal defect (VSD).
4. Right ventricular hypertrophy (enlargement of the right ventricle). The combined effect of these defects results in a right-to-left shunt, which leads to cyanosis.

🌀 Transposition of the Great Arteries (TGA): is a serious congenital heart defect in which the two main arteries leaving the heart—the aorta and the pulmonary artery—are switched. As a result, oxygen-poor blood is pumped to the body and oxygen-rich blood is circulated back to the lungs. Newborns with TGA often develop cyanosis (bluish skin) soon after birth and usually require surgery to correct the defect.

🌀 Persistent Truncus Arteriosus: Is where the truncal ridges and aortopulmonary septum

fail to divide the truncus into the aorta and pulmonary trunk, resulting in a single arterial vessel straddling a VSD.

4. Vascular Anomalies

✧ Coarctation of the Aorta: A congenital narrowing of the aorta that restricts blood flow and increases blood pressure.

✧ Persistent Left SVC: A congenital condition in which the left superior vena cava persists after birth instead of disappearing during fetal development. It is usually asymptomatic and found incidentally.

✧ Double Aortic Arch: This occurs when the right dorsal aorta does not disappear, resulting in the formation of an aortic arch on both sides. This forms a vascular ring around the trachea and esophagus, which can cause difficulty in breathing and swallowing.

✧ Persistent Atrioventricular Canal: Caused by the failure of the dorsal and ventral AV cushions to fuse, resulting in a large central opening in the heart and a common AV valve.